

## Unit 117 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 82.2 ac. 44 plots, BAF 20, variable-radius cruise on a 300 ft grid, 490 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

**TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE**

| STAND DENSITY        |                                        | STAND INFO          |                                                                                                                                         |
|----------------------|----------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Basal area           | 223 sq ft/ac (1 SE of mean, BA 2.0 %)  | Type                | Sierran mixed conifer                                                                                                                   |
| Trees per acre       | 89                                     | CWHR size / density | 4 / D (cover 88 %)                                                                                                                      |
| QMD                  | 21.4 in                                | SDI                 | 286 (49 % of max 580)                                                                                                                   |
| SAWTIMBER (>= 10 in) |                                        | BIOMASS (< 10 in)   |                                                                                                                                         |
| BA   QMD             | 219 sq ft/ac   22.4 in                 | BA   QMD            | 4 sq ft/ac   8.7 in                                                                                                                     |
| TPA                  | 80                                     | TPA                 | 9                                                                                                                                       |
| Volume               | 35.0 MBF/ac (6963 cu ft)               | Volume              | 1.3 green tons/ac (59 cu ft)                                                                                                            |
| TREATMENT TARGET     |                                        | CRUISE DESIGN       |                                                                                                                                         |
| Leave                | 35 % of max SDI, about 158 sq ft/ac BA | Vol. sampling error | 7.7 % at 95 % (CV 25 %, t = 2.02, n = 44)                                                                                               |
| Species (BA)         | WF 69, PP 66, IC 36, DF 33, SP 18      | Stratum standard    | 30 % of volume (R5 2409.12-2021-4, sec. 41.1): <b>MEETS</b> ; plots for standard 3                                                      |
|                      |                                        | Sale volume share   | 4.7 % of sale volume: standard shown for information (under the R5 10 % threshold)                                                      |
|                      |                                        | BA sampling error   | 4.0 % at 95 % (CV 13 %); reference, not the standard                                                                                    |
|                      |                                        | Measured trees      | PP 146, WF 151, DF 73, SP 40, IC 80; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): <b>ok</b> |
|                      |                                        | Plot count          | 44; R5 20-plot minimum for an area-based cruise (sec. 41.3): <b>ok</b>                                                                  |

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise\_data.xlsx, Standards and assumptions.

**TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE**

| DBH class, in | Trees      | TPA         | BA           | cu ft       | MBF         |
|---------------|------------|-------------|--------------|-------------|-------------|
| 4-7.9         | 1          | 1.4         | 0.5          | 6           | 0.0         |
| 8-11.9        | 20         | 16.0        | 9.1          | 166         | 0.8         |
| 12-15.9       | 40         | 17.3        | 18.2         | 392         | 2.0         |
| 16-19.9       | 67         | 17.1        | 30.5         | 763         | 3.8         |
| 20-23.9       | 83         | 14.2        | 37.7         | 1077        | 5.4         |
| 24-27.9       | 71         | 8.9         | 32.3         | 1044        | 5.2         |
| 28-31.9       | 66         | 6.2         | 30.0         | 1036        | 5.2         |
| 32-35.9       | 50         | 3.6         | 22.7         | 811         | 4.1         |
| 36-39.9       | 31         | 1.8         | 14.1         | 544         | 2.7         |
| 40-43.9       | 21         | 1.0         | 9.5          | 397         | 2.0         |
| 44-47.9       | 19         | 0.7         | 8.6          | 371         | 1.9         |
| 48-51.9       | 9          | 0.3         | 4.1          | 168         | 0.8         |
| 52-55.9       | 7          | 0.2         | 3.2          | 137         | 0.7         |
| 56-59.9       | 3          | 0.1         | 1.4          | 68          | 0.3         |
| 60-63.9       | 2          | 0.0         | 0.9          | 42          | 0.2         |
| <b>All</b>    | <b>490</b> | <b>89.0</b> | <b>222.7</b> | <b>7022</b> | <b>35.3</b> |

**TABLE 3. STOCK TABLE BY SPECIES, PER ACRE**

| Species    | Trees      | TPA         | BA           | QMD         | cu ft       | MBF         |
|------------|------------|-------------|--------------|-------------|-------------|-------------|
| WF         | 151        | 26.6        | 68.6         | 21.8        | 2191        | 11.0        |
| PP         | 146        | 28.4        | 66.4         | 20.7        | 2289        | 11.5        |
| IC         | 80         | 15.3        | 36.4         | 20.8        | 965         | 4.8         |
| DF         | 73         | 12.5        | 33.2         | 22.0        | 986         | 4.9         |
| SP         | 40         | 6.1         | 18.2         | 23.3        | 591         | 3.0         |
| <b>All</b> | <b>490</b> | <b>89.0</b> | <b>222.7</b> | <b>21.4</b> | <b>7022</b> | <b>35.3</b> |

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 117 plots, 300 ft grid (orange: QA flag)

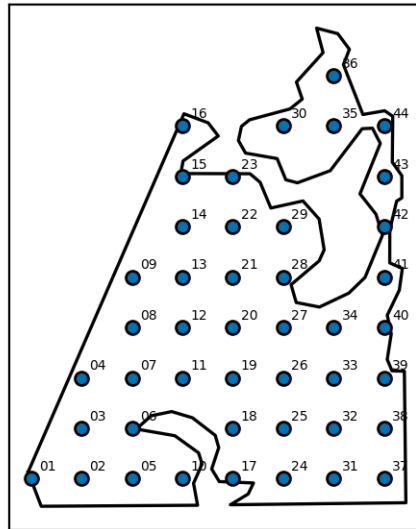


Figure 1. Plot locations with QA status.

**QA findings:** none; unit accepted

TABLE 4. PLOT LIST

| Plot   | Trees in | BA sq ft/ac | Slope % | Status |
|--------|----------|-------------|---------|--------|
| 117-01 | 11       | 220         | 23      | ok     |
| 117-02 | 12       | 240         | 37      | ok     |
| 117-03 | 13       | 260         | 28      | ok     |
| 117-04 | 13       | 260         | 9       | ok     |
| 117-05 | 10       | 200         | 22      | ok     |
| 117-06 | 10       | 200         | 39      | ok     |
| 117-07 | 11       | 220         | 23      | ok     |
| 117-08 | 12       | 240         | 30      | ok     |
| 117-09 | 9        | 180         | 22      | ok     |
| 117-10 | 11       | 220         | 21      | ok     |
| 117-11 | 12       | 240         | 14      | ok     |
| 117-12 | 11       | 220         | 15      | ok     |
| 117-13 | 13       | 260         | 12      | ok     |
| 117-14 | 12       | 240         | 18      | ok     |
| 117-15 | 12       | 240         | 35      | ok     |
| 117-16 | 10       | 200         | 27      | ok     |
| 117-17 | 9        | 180         | 30      | ok     |
| 117-18 | 11       | 220         | 20      | ok     |
| 117-19 | 11       | 220         | 25      | ok     |
| 117-20 | 12       | 240         | 25      | ok     |
| 117-21 | 12       | 240         | 11      | ok     |
| 117-22 | 13       | 260         | 9       | ok     |
| 117-23 | 10       | 200         | 30      | ok     |
| 117-24 | 11       | 220         | 34      | ok     |
| 117-25 | 6        | 120         | 19      | ok     |
| 117-26 | 11       | 220         | 27      | ok     |
| 117-27 | 13       | 260         | 36      | ok     |
| 117-28 | 12       | 240         | 22      | ok     |
| 117-29 | 12       | 240         | 19      | ok     |
| 117-30 | 13       | 260         | 25      | ok     |
| 117-31 | 10       | 200         | 33      | ok     |
| 117-32 | 10       | 200         | 21      | ok     |
| 117-33 | 12       | 240         | 22      | ok     |
| 117-34 | 11       | 220         | 10      | ok     |
| 117-35 | 12       | 240         | 22      | ok     |
| 117-36 | 11       | 220         | 26      | ok     |
| 117-37 | 11       | 220         | 36      | ok     |
| 117-38 | 13       | 260         | 17      | ok     |
| 117-39 | 12       | 240         | 15      | ok     |
| 117-40 | 12       | 240         | 33      | ok     |
| 117-41 | 9        | 180         | 18      | ok     |
| 117-42 | 9        | 180         | 33      | ok     |
| 117-43 | 11       | 220         | 27      | ok     |
| 117-44 | 9        | 180         | 28      | ok     |